



Morning Brief

Ludovici · Friday, June 19, 2026

In this edition

AI Governance & Policy · Cybersecurity & Threats · On the Federal Front · AI Industry Landscape · Project Status · Five Project Briefs · Training · Signal & Trajectory · Week in Review

By Stephanie Ludovici

A personal learning and awareness brief on AI governance, cybersecurity, the federal landscape, and the science behind the methods. Every news item below is sourced; analysis is labeled as such.

AI Governance & Policy

- **NSPM-11 makes AI in the national security enterprise its own governing regime.** Signed June 5, the National Security Presidential Memorandum directs the national security enterprise, the Department of War, and the Intelligence Community to accelerate AI adoption "with rigorous oversight," rescinds the Biden-era NSM-25, and tasks the Committee on National Security Systems and OMB — within 90 days — to issue AI-in-NSS governance policy aligned to OMB M-25-21, plus a joint DNI/DoW/NSA push on AI risk management, baseline AI security practices, and standardized AI test, evaluation, verification & validation (TEVV). [White House Fact Sheet](#) · [CFR analysis](#) — *Why it matters:* this is the new charter for AI in national security systems, and its standardized-TEVV and assurance mandate is exactly the federal/IC anchor your governance work maps controls against.

- **NIST drops "safety" from its AI consortium and refocuses it on measurement and evaluation.** Per a May 29 Federal Register notice (reported June 1), the former AI Safety Institute Consortium is now the NIST Artificial Intelligence Consortium, with ~280 members reorganized into six task groups centered on AI measurement, an evaluation ecosystem, and adoption — including a documentation-cards group standardizing templates for datasets, models, systems, and **AI TEVV**, and a BENGAL group studying LLM use for **intelligence analysis**. [NIST](#) · [FedScoop](#) — *Why it matters:* this is where the federal AI evaluation/TEVV artifacts you map to will actually be drafted, and the intelligence-analysis task group is squarely in your lane.
- **NSPM-12 rebuilds the cybersecurity chain of command that NSPM-11's AI rules will run on.** Signed June 12, it re-establishes the Committee on National Security Systems (updated for the first time in 35+ years), designates the NSA Director as National Manager for NSS cybersecurity with authority to issue binding baseline requirements and emergency directives, and rescinds NSD-42 (1990) and NSM-8 (2022). [White House](#) · [SecurityWeek](#) — *Why it matters:* it names who owns NSS cyber governance — the same CNSS/NSA structure NSPM-11 tasks with writing AI-in-NSS policy, so the two memos are best read together.

Cybersecurity & Threats

- **The first Splunk flaw ever on CISA's KEV list triggered the first BOD 26-04 three-day sprint.** CVE-2026-20253 (CVSS 9.8) is an unauthenticated arbitrary-file-write bug in a Splunk Enterprise PostgreSQL sidecar endpoint that researchers chained to remote code execution; it hits Splunk Enterprise 10.2 < 10.2.4 and 10.0 < 10.0.7 (patched June 10). CISA added it June 18 and gave FCEB agencies until **June 21** under the new risk-based directive. [SecurityWeek](#) · [CISA alert](#) — *Federal relevance:* Splunk is a core federal SIEM; a SIEM compromise hits the very telemetry defenders rely on, and this is BOD 26-04's first live test.
- **A maximum-severity Joomla editor flaw is under active exploitation.** CVE-2026-48907 (CVSS 10.0), an improper-access-control bug in the JCE "Widget Factory" content editor, lets unauthenticated users create editor profiles and upload/execute PHP; fixed in JCE 2.9.99.5. CISA added it to KEV June

16 with an FCEB due date of June 19. [The Hacker News](#) · [CISA alert](#) — *Federal relevance*: CMS plugins remain a soft underbelly of public-facing .gov sites; unauthenticated RCE on a content editor is exactly the "public exposure × automatable" profile BOD 26-04 prioritizes.

- **Attackers are hitting FortiSandbox bugs Fortinet quietly patched in April.** CVE-2026-39808 (OS-command injection) and CVE-2026-39813 (path traversal) are being exploited in the wild — VulnCheck dated the first activity to June 9, and Defused logged dozens of attempts from multiple independent operators across several countries. [CyberScoop](#) — *Federal relevance*: FortiSandbox is a trusted malware-analysis appliance widely deployed in government; turning a security tool into an entry point is the worst kind of supply-chain inversion.
- **Cisco faces its seventh actively exploited SD-WAN zero-day of the year — with no patch yet.** CVE-2026-20245, a validation-error flaw in Catalyst SD-WAN Manager flagged by Mandiant, can run commands as root and is reachable by first abusing earlier zero-days; Cisco says no fix or workaround is available. (Distinct from the earlier Catalyst SD-WAN KEV CVE-2026-20262.) [CyberScoop](#) — *Federal relevance*: SD-WAN management planes are high-value pivots in federal networks, and an unpatched root-level zero-day keeps that risk open.

On the Federal Front

- **A senator is pressing the Pentagon over the rushed rewrite of its autonomous-weapons policy.** In a letter reported June 15, Sen. Ruben Gallego (D-Ariz.) questioned whether the compressed 90-day timeline to revise DoD Directive 3000.09 — mandated by NSPM-11 — will erode testing, safeguards, and human-judgment requirements, and asked the Secretary of War to answer a list of questions by June 26. [DefenseScoop](#) — *Why it matters*: it is the sharpest congressional oversight yet on how DoW will govern AI-enabled targeting as NSPM-11 strips away prior oversight language — the human-oversight question your framework scores, at the highest stakes.
- **The Senate's FY2027 NDAA would create a single Pentagon owner for IT, cyber, and AI.** SASC's draft proposes a new Under Secretary of Defense for Cyber, Information, and Networks — dual-hatted as DoD CIO and Principal Cyber

Advisor — with the Chief Digital and AI Office (CDAO) nested under it; the House bill directs a reorganization review but stops short of a new under secretary. [DefenseScoop](#) — *Why it matters*: consolidating AI and cyber accountability under one senior official would reshape how the department governs data and AI-enabled operations — watch this in NDAA conference.

- **GAO faults the DoD/VA electronic-health-record program for missing cyber and privacy measures.** Report GAO-26-107673 (June 2) found the Federal Electronic Health Record Modernization office lacked interagency cybersecurity/privacy goals and performance measures for two fiscal years even as the system (Oracle Health) heads toward 500,000+ users serving 18 million people; DoD did not concur with the recommendations. [FedScoop](#) — *Why it matters*: it is a concrete accountability gap around one of the government's largest pools of sensitive personal data — a data-governance failure mode, not just an IT one.
- **DISA's JWCC successor would make AI, edge, and cross-domain data sharing the baseline.** A draft performance-of-work statement (reported June 1) for the "Unified Cloud Marketplace" requires vendors to deliver AI/ML services across all classification and impact levels — including denied/degraded environments — plus secure cross-domain transfer and NSA-certified red-team exercises, with a final RFP expected by year's end. [DefenseScoop](#) — *Why it matters*: the successor to the \$9B JWCC vehicle would set the security, AI, and data-sharing governance floor for how the entire department buys commercial cloud and AI.

AI Industry Landscape

- **Anthropic opened a Seoul office and tied it to public-sector AI safety.** On June 17 the company launched its third Asia-Pacific office alongside Korean enterprise deployments (NAVER, Samsung SDS, LG CNS) and a research-access program; trade coverage reports an accompanying memorandum of understanding with Korea's science ministry covering public-sector adoption, Korean-language model safety evaluations with the Korea AI Safety Institute, and information-sharing on AI-enabled cyber threats. [Anthropic](#) · [Digital Watch](#) — *Why it matters*: U.S. labs are racing to lock in allied-government AI-

safety and cyber partnerships abroad even as Washington tightens export controls on those same frontier models.

- **At the G7, the major AI CEOs pitched a U.S.-led forum to set global standards for frontier models.** Over a June 17 working lunch in Évian, OpenAI's Sam Altman floated an international standards forum that Anthropic's Dario Amodei and Google DeepMind's Demis Hassabis backed as a U.S.-led coalition; one leader likened it to the Financial Stability Board — days after Washington shut down foreign access to Anthropic's Mythos and Fable models. [CNBC](#) · [Semafor](#) — *Why it matters:* industry is lobbying heads of state to shape frontier-model governance directly, positioning a democratic-bloc standards body against both Chinese AI and unilateral export controls.
- **OpenAI says ChatGPT goes live on the Pentagon's GenAI.mil in "early July."** Speaking at the Defense One Tech Summit on June 16, an OpenAI executive said ChatGPT will reach 3 million+ defense personnel through the CDAO platform, certified for Controlled Unclassified Information at Impact Level 5; GenAI.mil already reports 1.3M+ users and 100,000+ built agents. [Defense One](#) · [Nextgov/FCW](#) — *Why it matters:* a frontier commercial chatbot is about to enter routine Pentagon workflows at scale, raising the stakes for TEVV, groundedness, and provenance controls to precede fielding, not follow it.
- **Google released Gemma 4 12B, an open-weight model that runs on a 16GB laptop.** Announced June 3 under Apache 2.0, the multimodal model uses an encoder-free architecture for native vision/audio and approaches Google's larger MoE on benchmarks at under half the memory, shipping on Hugging Face, Ollama, and LM Studio. [Google](#) — *Why it matters:* capable open weights running offline complicate the very export-control and frontier-gating regime Washington is building, because open models can't be remotely disabled the way Fable 5 was.

Project Status

- **Archive Project (Archive Discovery & Transcription Assistant)** (*newest — created today, June 19*). A portfolio-grade, agentic POC on public data: handwritten-text recognition, agentic natural-language search with cited results, and cross-document entity/relationship discovery — built Library-of-

Congress-first (*By the People* / Concordia, open API) and source-agnostic so the NARA Catalog plugs in second. **Next action:** ship the leanest defensible slice — HTR + an entity graph on one completed LoC campaign — and score the HTR against LoC's published human-verified transcription CSVs as day-one ground truth.

- **work-newsletter ("The Liberty Brief")** (*modified today, June 19*). The sibling daily-brief generator (`build_edition.py` touched this morning) that builds an edition from a template with live research and emits HTML/PDF/notebook. **Next action:** factor out the shared research/ledger logic so it and this morning brief don't double-cover the same governance/cyber items — one novelty index, two outputs.
- **NetMon-Server** (*modified June 15*). Single-file PowerShell loopback SIEM/monitor with built-in cmdlets, DPAPI-encrypted storage, and an optional on-device AI layer. **Next action, tied to today's news:** add a KEV cross-reference view so KEV-listed CVEs on monitored/exposed hosts surface first — today's Splunk CVE-2026-20253 (with its June 21 BOD 26-04 deadline) and Joomla CVE-2026-48907 are the live examples.
- **AI Framework (v3.0 roadmap tie-in)**. NSPM-11's standardized-TEVV/assurance mandate and the renamed NIST AI Consortium's measurement/TEVV and documentation-cards work are the non-OWASP, federal-and-standards triangulation the v3.0 roadmap says the agentic and groundedness overlays need before any control is drafted. **Next action:** log NSPM-11 (TEVV/assurance) and the NIST AI Consortium task groups into the v3.0 evidence base as TEVV anchors, alongside the existing OWASP and NIST agentic-identity threads.

Five Project Briefs

Full briefs also saved to `project-ideas/2026-06-19-ideas.md`.
None repeats a concept in the ledger.

1. Diffusion posterior sampling for sparse-view CT reconstruction

- **Domain:** Computational / medical imaging, inverse problems.
Language(s): Python (PyTorch).
- **Problem statement.** Reducing CT dose means fewer projection angles, but filtered back-projection then streaks and analytic regularizers blur detail. A learned generative prior over clean

images can fill the missing-angle information *and* return a posterior, so the reconstruction carries its own uncertainty.

- **Objective(s).** Train a score-based diffusion prior on clean CT slices, reconstruct from sparse-view sinograms by diffusion posterior sampling, and quantify quality and uncertainty vs. filtered back-projection and a total-variation baseline.
- **Questions to answer.**
 - How does quality (PSNR/SSIM) scale with view count (180 → 60 → 30 → 15) for diffusion posterior sampling vs. FBP and TV?
 - Does the per-pixel posterior spread actually track reconstruction error (calibrated uncertainty)?
 - How sensitive is the result to the data-consistency step size and noise assumption?
- **Proposed method/approach.** Train an unconditional score model on clean slices; at inference, alternate reverse-diffusion steps with a measurement-consistency gradient through the differentiable Radon operator (DPS); draw multiple posterior samples for mean and variance; evaluate across view counts with known-truth injection.
- **Tools.** PyTorch, diffusers, a differentiable Radon transform (tomosipo/ASTRA or torch-radon; scikit-image radon for a CPU prototype), NumPy/matplotlib. **Datasets:** LoDoPaB-CT or the AAPM Low-Dose CT Grand Challenge; Shepp–Logan phantoms for a fast first pass. **Compute:** one 8–12 GB GPU (CPU for the phantom prototype).
- **Scope.** In: 2-D slices, one anatomy, three view counts, posterior-spread calibration. Out: 3-D volumetric reconstruction, clinical deployment, metal-artifact correction. **Deliverables:** trained score model, quality-vs-views curves, uncertainty-calibration plot, reproducible notebook.
- **Estimated duration to complete.** 5–7 weeks part-time.
- **Sources.** Chung et al., *Diffusion Posterior Sampling for General Noisy Inverse Problems*, ICLR 2023 — <https://arxiv.org/abs/2209.14687> · Song et al., *Score-Based Generative Modeling through SDEs*, ICLR 2021 — <https://arxiv.org/abs/2011.13456>

2. PCMC1+ causal discovery for climate teleconnections

- **Domain:** Climate science / causal discovery. **Language(s):** Python (tigramite).

- **Problem statement.** Climate indices (ENSO, NAO, PDO, AO) are correlated through shared drivers and lagged influence, so plain correlation networks conflate direct, indirect, and common-cause links. Causal-discovery methods built for autocorrelated, high-dimensional time series can recover a defensible lagged causal graph that correlation cannot.
- **Objective(s).** Reconstruct a lagged and contemporaneous causal graph among major climate indices with PCMCi+, test its stability, and compare it against the correlation network and known physical teleconnections.
- **Questions to answer.**
 - Which directed, time-lagged links survive conditional-independence testing, and at what lags?
 - How do recovered links change with the independence test (linear vs. nonlinear) and the maximum lag?
 - Do the discovered links match documented teleconnection physics, and where do they diverge?
- **Proposed method/approach.** Assemble monthly standardized index series; check stationarity; run PCMCi+ with both linear (ParCorr) and nonlinear (GPDC/CMIknn) tests across a lag range; assess robustness via bootstrap and subsampling; contrast the time-series graph with a thresholded correlation graph.
- **Tools.** Python, tigramite, statsmodels, pandas, NetworkX/matplotlib. **Datasets:** NOAA PSL climate indices (Niño 3.4, NAO, PDO, AO); optionally ERA5 for an extension. **Compute:** laptop CPU.
- **Scope.** In: ~5–8 indices, lagged + contemporaneous discovery, test/lag sensitivity, physics cross-check. Out: gridded field-level discovery, mechanistic attribution, forecasting. **Deliverables:** causal-graph figures, robustness report, correlation-vs-causal comparison, reproducible script.
- **Estimated duration to complete.** 3–4 weeks part-time.
- **Sources.** Runge et al., *Detecting and quantifying causal associations in large nonlinear time series datasets*, Science Advances 2019 — <https://doi.org/10.1126/sciadv.aau4996> · Runge, *Discovering contemporaneous and lagged causal relations* (PCMCi+), UAI 2020 — <https://proceedings.mlr.press/v124/runge20a.html>

3. Echo State Networks (reservoir computing) for chaotic spatiotemporal forecasting

- **Domain:** Nonlinear dynamics / time-series forecasting.
Language(s): Julia (`ReservoirComputing.jl`).
- **Problem statement.** Forecasting chaotic systems is hard and training recurrent nets is expensive. Reservoir computing fixes a large random recurrent reservoir and trains only a linear readout — so it learns in seconds on a laptop and is a strong baseline for short-horizon chaos prediction. Julia's reservoir tooling makes the method first-class.
- **Objective(s).** Build an Echo State Network that forecasts a chaotic system, measure the valid prediction horizon in Lyapunov times, and study how reservoir size, spectral radius, and regularization control accuracy and stability.
- **Questions to answer.**
 - How far ahead (in Lyapunov times) can the ESN forecast before trajectories diverge, vs. a naive baseline?
 - How do spectral radius, leak rate, and ridge penalty trade off one-step error against free-running stability?
 - Does the trained readout reproduce the system's long-run attractor statistics, not just short-term tracks?
- **Proposed method/approach.** Generate Lorenz-63 and Mackey–Glass trajectories; split into washout/train/test; fit the linear readout by ridge regression; evaluate teacher-forced and free-running error; sweep hyperparameters; compare the free-run attractor's invariant statistics to truth.
- **Tools.** Julia: `ReservoirComputing.jl`, `DifferentialEquations.jl`, `Statistics`, `Plots.jl`.
Datasets: self-generated Lorenz-63 / Mackey–Glass (plus optional sunspot series). **Compute:** laptop CPU.
- **Scope.** In: two canonical systems, valid-time and attractor-statistic metrics, hyperparameter sweep. Out: large spatiotemporal PDE systems, deep/learned reservoirs, real-time control. **Deliverables:** ESN implementation, valid-time curves, attractor-statistic comparison, Julia notebook.
- **Estimated duration to complete.** 2–4 weeks part-time.
- **Sources.** Jaeger & Haas, *Harnessing Nonlinearity*, Science 2004 — <https://doi.org/10.1126/science.1091277> · Pathak et al., *Model-Free Prediction of Large Spatiotemporally Chaotic*

4. Calibrated survival analysis with random survival forests and competing risks

- **Domain:** Biostatistics / epidemiology. **Language(s):** R (randomForestSRC, riskRegression).
- **Problem statement.** Time-to-event data have censoring and often competing risks, which classification and even Cox models can mishandle. Random survival forests capture nonlinearity without a proportional-hazards assumption — but a model is only useful if its predicted risks are *calibrated* and its discrimination holds out of sample.
- **Objective(s).** Fit random survival forests (and a competing-risks variant) on a public cohort, then rigorously evaluate discrimination and calibration against a Cox baseline using censoring-aware metrics.
- **Questions to answer.**
 - Does the random survival forest beat Cox on time-dependent AUC and the Brier/IPA score, and at which horizons?
 - Are predicted absolute risks calibrated (calibration curves at fixed times), or systematically off?
 - Under competing risks, how different are cause-specific cumulative-incidence estimates from a naive Kaplan–Meier read?
- **Proposed method/approach.** Fit Cox and random survival forests; for competing risks fit a cause-specific RSF; evaluate time-dependent ROC/AUC, integrated Brier score and IPA via prediction-error curves, and calibration plots at chosen horizons; validate with repeated cross-validation; sanity-check variable importance.
- **Tools.** R: survival, randomForestSRC, riskRegression, pec, prodlim, tidyverse. **Datasets:** public R cohorts (survival::rotterdam, gbsg, pbc); a competing-risks set (survival::mgus2) for the extension. **Compute:** laptop CPU.
- **Scope.** In: one cohort, RSF vs. Cox, discrimination + calibration + competing risks, cross-validated. Out: deep survival nets, external multi-cohort validation, individual clinical claims.
Deliverables: model-comparison table, calibration/Brier curves, cumulative-incidence figures, reproducible R script.

- **Estimated duration to complete.** 3–4 weeks part-time.
- **Sources.** Ishwaran et al., *Random Survival Forests*, Annals of Applied Statistics 2008 — <https://doi.org/10.1214/08-AOAS169> · Gerds & Schumacher, *Consistent estimation of the expected Brier score* — <https://doi.org/10.1002/bimj.200610047>

5. Conditional Neural Processes for amortized ocean field reconstruction from sparse sensors

- **Domain:** Physical oceanography / spatial machine learning.
Language(s): Python (PyTorch).
- **Problem statement.** Reconstructing a continuous field (e.g., sea-surface temperature) from sparse, irregular measurements is the classic interpolation problem, usually solved with kriging that must be re-fit per map. Neural processes *amortize* this: one trained model maps any set of observations to a predictive field with uncertainty in a single forward pass — fast, and handling varying sensor counts and locations.
- **Objective(s).** Train a Conditional/Attentive Neural Process to reconstruct an SST field from sparse samples with calibrated uncertainty, and compare accuracy and speed against Gaussian-process kriging.
- **Questions to answer.**
 - Does the (Attentive) Neural Process match or beat kriging on held-out RMSE while being far faster across many maps?
 - Is the predicted uncertainty calibrated, and does it widen where observations are sparse?
 - How does skill degrade as the observation count drops (dense → very sparse)?
- **Proposed method/approach.** Build training tasks by masking gridded SST into sparse context/target splits; train a CNP, then an Attentive NP, with a Gaussian-likelihood (NLL) objective; evaluate held-out RMSE and interval coverage vs. a GP baseline; stress-test under shrinking context; optionally validate against true sparse Argo locations.
- **Tools.** Python: PyTorch, xarray, NumPy, matplotlib; scikit-learn/GPy for the GP baseline. **Datasets:** NOAA OISST (gridded SST); Argo float positions for a realistic sparse-sampling test. **Compute:** one consumer GPU (CPU feasible at coarse resolution).

- **Scope.** In: one region, SST, CNP vs. ANP vs. GP, accuracy + calibration + speed. Out: full-globe operational reanalysis, multivariate ocean state, dynamical data assimilation.
Deliverables: trained models, RMSE/coverage tables, reconstruction figures, speed benchmark, reproducible notebook.
- **Estimated duration to complete.** 4–6 weeks part-time.
- **Sources.** Garnelo et al., *Conditional Neural Processes*, ICML 2018 — <https://arxiv.org/abs/1807.01613> · Kim et al., *Attentive Neural Processes*, ICLR 2019 — <https://arxiv.org/abs/1901.05761>

Training

1. Python — `functools.singledispatch`

`singledispatch` turns a plain function into a generic one that selects an implementation based on the type of its first argument — a clean replacement for long `isinstance` ladders, and extensible because new types can be registered later, even by other modules.

```
# singledispatch picks an implementation based on the first argument
from functools import singledispatch

@singledispatch
def describe(value):
    # This base case runs when no registered type matches
    return f"unhandled type: {type(value).__name__}"

@describe.register
def _(value: int):
    # Registered for int; dispatch uses the runtime type, not the annotation
    return f"int with bit_length {value.bit_length()}"

@describe.register
def _(value: list):
    # Registered for list; the name "_" is conventional for unnamed functions
    return f"list of length {len(value)}"

print(describe(42))          # int with bit_length 6
print(describe([1, 2]))     # list of length 2
print(describe(3.14))       # falls back to the base implementation
```

Why it matters: it makes type-based behavior open for extension without editing the original function — the dispatch table lives outside it. **Common pitfall:** it dispatches on the *first* argument only, so it is the wrong tool for multi-argument dispatch; inside a class use `functools.singledispatchmethod`; and because it keys on the type's MRO, register an abstract base (e.g., `numbers.Integral`) when you want to catch all subtypes at once. **Reference:** Python docs, `functools.singledispatch` — <https://docs.python.org/3/library/functools.html#functools.singledispatch>

2. R — Leak-proof preprocessing with `recipes` (tidymodels)

A recipe declares preprocessing as steps whose parameters are *learned from training data only* (prep) and then *applied* unchanged to new data (bake). That separation is what prevents data leakage — the silent error where centering, scaling, or imputation uses statistics computed over the whole dataset (including the test set) and inflates measured performance.

```
# A recipe learns step parameters on training data, then
library(recipes)

rec <- recipe(mpg ~ ., data = mtcars) |>
  step_impute_median(all_numeric_predictors()) |> # median
  step_normalize(all_numeric_predictors()) |>    # centered
  step_dummy(all_nominal_predictors())          # encoded

prepped <- prep(rec, training = mtcars)          # estimate all
train_x <- bake(prepped, new_data = NULL)        # apply to the
# bake(prepped, new_data = test) applies the SAME learned
```

Why it matters: preprocessing is part of the model, so its parameters must be estimated inside each resample — `recipes` composes with `rsample/workflows` so this happens automatically during cross-validation. **Idiomatic note:** select columns by role or type (`all_numeric_predictors()`), not by hard-coded names, so the recipe survives schema changes. **Reference:** Kuhn & Silge, *Tidy Modeling with R* — <https://www.tmwr.org/>

3. Data-centric — Bitemporal data modeling

Most history-tracking keeps one time axis. Bitemporal modeling keeps **two**: *valid time* (when a fact is true in the real world) and *transaction/system time* (when the database recorded it). Keeping both lets you answer the audit-grade question "what did we believe on date X about

the state as of date Y" — and makes corrections additive (new rows) rather than destructive (overwrites).

```
-- Each row carries TWO time ranges: valid time (real world)
CREATE TABLE employee_title (
  employee_id INT,
  title        TEXT,
  valid_from   DATE,          -- when the title became true
  valid_to     DATE,          -- when it stopped being true
  sys_from     TIMESTAMP,     -- when this row was recorded
  sys_to       TIMESTAMP      -- when this row was superseded
);

-- "As of" query: what we believed on 2026-03-01 about the title
SELECT title FROM employee_title
WHERE employee_id = 7
      AND DATE '2026-01-15' BETWEEN valid_from AND valid_to
      AND TIMESTAMP '2026-03-01 00:00:00' BETWEEN sys_from AND sys_to;
```

Why it matters: it is the backbone of reproducible, defensible reporting — you can reconstruct exactly the number you would have reported on any past date, even after later corrections. It generalizes Slowly Changing Dimensions (which track a single axis). **Note:** SQL: 2011 standardizes application-time and system-time periods, and several engines now support them natively. **Reference:** Snodgrass, *Developing Time-Oriented Database Applications in SQL* (free PDF) — <https://www2.cs.arizona.edu/~rts/tddbbook.pdf>

4. AI method — Echo State Networks (reservoir computing)

An Echo State Network is a recurrent network with a twist: a large, *fixed*, randomly wired reservoir supplies a high-dimensional nonlinear memory of the input history, and **only a linear readout is trained**. There is no backpropagation through time.

Intuition/math. With fixed input weights W_{in} and fixed recurrent weights W , the state updates as $x_t = \tanh(W_{in} u_t + W x_{t-1})$ (optionally with a leak rate). W is rescaled to a *spectral radius* below ~ 1.25 so the reservoir has fading memory (the "echo-state property"). The only trained part is the readout W_{out} , fit by ridge regression mapping collected states to targets — a single linear solve, hence seconds of training. The reservoir behaves like a fixed nonlinear temporal feature map; the readout reads off what it needs.

When to use it. Short-horizon forecasting of nonlinear or chaotic dynamics, low-data/low-compute settings, and as a strong baseline before reaching for deep RNNs or transformers. Tune reservoir size,

spectral radius (memory vs. stability), input scaling, leak rate, and the ridge penalty. Because chaotic systems have a finite predictability horizon, report it in Lyapunov times and check attractor statistics, not just pointwise error. Demonstrated in today's companion notebook (notebooks/2026-06-19-echo-state-networks.ipynb) on Lorenz-63, with explicit M&S (simulate + train) and T&E (one-step RMSE vs. persistence; free-running valid time in Lyapunov times).

Reference: Jaeger & Haas, *Harnessing Nonlinearity*, Science 2004 — <https://doi.org/10.1126/science.1091277>

Signal & Trajectory

Analysis and inference, not reported fact. Each bullet is grounded in an item cited above.

- **National-security AI governance is consolidating into one AI-plus-cyber chain of command.** If NSPM-11, NSPM-12, and the SASC under secretary proposal (all above) hold, expect AI-in-NSS rules to be written by CNSS/NSA with standardized TEVV baked in; watch the 90-day NSPM-11 deliverable and the NIST AI Consortium's documentation/TEVV task groups for the first concrete artifacts you would map controls to.
- **Risk-based patching is now operational, not theoretical.** With Splunk CVE-2026-20253 becoming BOD 26-04's first three-day deadline (June 21) atop the Joomla and FortiSandbox exploitation (above), expect KEV-plus-exploitability to set federal patch cadence; watch whether the Splunk deadline becomes the enforcement template — and stand up the KEV cross-reference view noted for NetMon-Server.
- **Commercial frontier AI is entering DoD at scale while the U.S. gates models abroad.** If OpenAI's early-July GenAI.mil launch and Anthropic's Korea partnership (above) both land, expect a widening gap between rapid internal adoption and external control; watch whether TEVV, groundedness, and provenance controls actually precede fielding to 3M+ users, because retrofitting them is harder.
- **The standards-setting venue is moving.** If the G7's proposed U.S.-led forum and the rebranded NIST AI Consortium converge (above), expect evaluation/measurement standards to be set in industry-government venues rather than pure regulation; watch the consortium's task groups, since whatever "AI documentation

card" and TEVV templates they publish become the de facto control map.

Week in Review

Friday capstone. The daily Signal & Trajectory above stays focused on today; this synthesizes the week of June 15–19, grounded in the editions actually published (No. 5–No. 9).

Recap — what actually moved this week.

- **Agentic-AI governance shifted from advice toward operational pressure.** The week opened with the OWASP production-failure finding and two exploited agentic supply-chain flaws (LangGraph RCE; the Claude Code GitHub Action prompt-injection) and NIST's agent-identity work (Mon–Thu), and closed with NSPM-11's standardized-TEVV/assurance mandate and the NIST AI Consortium's measurement refocus (today) — guardrails visibly racing deployment.
- **Federal patching hard-pivoted to risk and exploitability.** CISA's BOD 26-04 (covered Thursday) got its first live invocation today on Splunk CVE-2026-20253 with a three-day deadline, capping a week of KEV additions (Oracle PeopleSoft, Cisco SD-WAN, LiteSpeed, and today's Joomla 10.0).
- **National-security AI got a new charter and a new chain of command.** NSPM-11 (AI in the national security enterprise) and NSPM-12 (CNSS re-establishment; NSA as National Manager), plus the SASC proposal to consolidate IT, cyber, and CDAO under one under secretary, together redraw who owns AI and cyber governance in NSS.
- **The government-AI market reshaped around access and control.** The \$1 access offers (covered Thursday), today's OpenAI→GenAI.mil and Anthropic→Korea moves, the national-security cohort that still excludes Anthropic and the BIS Fable/Mythos suspension (covered Tue/Wed), and the G7 push for a U.S.-led standards forum (today) are all facets of one contest: scale adoption fast, gate the most capable models tightly.

Trajectory ahead — framed conditionally.

- If NSPM-11's 90-day clock holds, expect a draft AI-in-NSS governance policy and CNSS standup by early fall; watch for

standardized AI TEVV requirements cascading into IC authorizations.

- If BOD 26-04 enforcement sticks, expect more three-day remediation sprints; the Splunk June 21 deadline is the one to watch as the template.
- If SASC's under secretary survives NDAA conference, DoD AI-and-cyber accountability consolidates under a single official — a structural change worth tracking against the House's lighter language.
- If the G7 forum and the renamed NIST AI Consortium converge, expect concrete evaluation standards to emerge before — or lagging behind — the commercial models (GenAI.mil) already scaling into use.

Ledgers updated: publication-ledger.md (No. 9), project-ideas/_ledger.md (5 ideas), and training-ledger.md (4 topics).

Attached: [brief \(.pdf\)](#) + [companion notebook \(.ipynb\)](#) + [project briefs \(.docx\)](#)

Morning Brief · Ludovici · Friday, June 19, 2026
By Stephanie Ludovici